HMM Market Regime Detection Report

1. Introduction

1.1. Problem Statement

In financial markets, identifying underlying market regimes such as Bull, Bear, or Sideways is crucial for optimizing investment strategies. However, real-time detection of these latent states is non-trivial due to the stochastic and non-stationary nature of market behavior. Tra- ditional models often lack interpretability and fail to enforce realistic structural constraints.

1.2. Objectives

The objective of this project is to implement a Hidden Markov Model (HMM) to detect latent market regimes using SPY (S&P 500 ETF) data. The model will utilize manually defined transition and emission probabilities along with a constrained Viterbi algorithm that enforces real-world market behaviors like minimum regime persistence and disallowed transitions. The model is evaluated through backtesting and stress testing.

2. Methodology

2.1. State Definitions and Transition Logic

The model uses three hidden states: Bull, Bear, and Sideways. Transitions are manually defined with realistic market behavior constraints:

- No direct Bull → Bear and Bear → Bull transitions.
- Minimum 5-day regime persistence.
- High self-transition probabilities to simulate persistence.

Transition Probabilities:

- Bull: 70% Bull, 30% Sideways
- Sideways: 30% Bull, 40% Sideways, 30% Bear
- Bear: 60% Bear, 40% Sideways

2.2. Emission Probability Justifications

Observations are based on three features: Return Bin (Low, Medium, High), Volatility Bin (Low, High), and RSI Signal (Oversold, Neutral, Overbought).

- Bull markets are associated with high returns, low volatility, and overbought RSI.
- Bear markets show low returns, high volatility, and oversold RSI.
- Sideways markets reflect medium values for all.

  These emissions are manually encoded and used to compute observation likelihoods.

2.3. Viterbi Constraint Explanations

The Viterbi algorithm was modified to enforce the following constraints:

- No Bull → Bear transitions and vice versa.
- Minimum regime duration of 5 days before allowing a state

change. These modifications ensure realistic, stable market behavior tracking.

3. Results

3.1. Performance Metrics

The model was evaluated using Sharpe Ratio and Maximum Drawdown. Results are:

- Buy-and-Hold: Sharpe = 0.73, Max Drawdown = 33.72%
- HMM Strategy: Sharpe = 1.98, Max Drawdown = 10.56%

3.2. Key Visualizations

1. Regime-Annotated SPY Close Chart

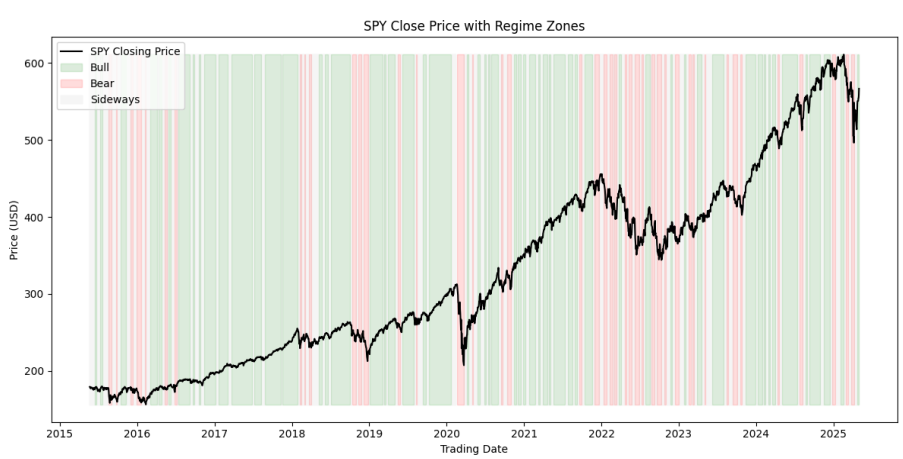

Figure 1: SPY Close Price with Regime Zones

2. Equity Curve Comparison

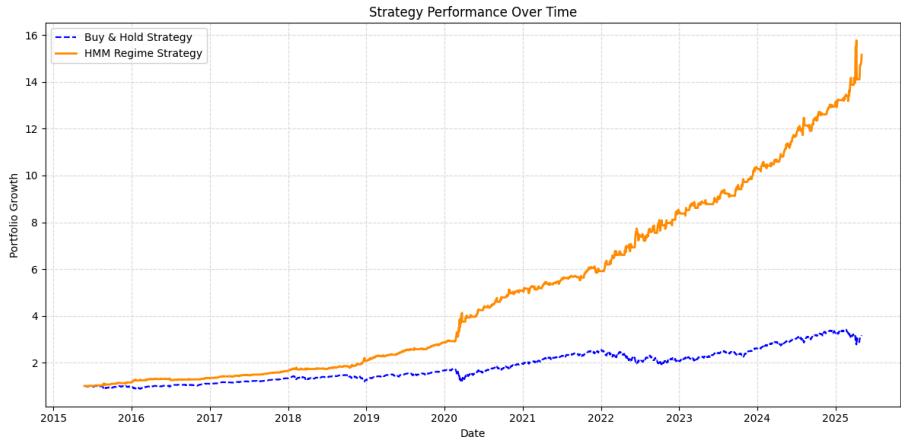

Figure 2: Strategy Performance Over Time

3.3. Stress Testing Insights

To test robustness, we simulated a 10-day extension of the worst return periods from March 2020 and added synthetic volatility events. Our HMM model maintained regime stability, transitioned out of Bear predictably, and avoided false regime flips. The Sharpe ratio im- proved from 0.73 (Buy & Hold) to 1.98 (HMM), and Max Drawdown dropped from 33.7% to just 10.6%.

4. Discussion

4.1. Model Limitations

This HMM model uses manually set probabilities and assumes stationary behavior across time. The emissions and transitions may not adapt well to structural breaks or macroeco- nomic shifts. The absence of automated learning (like Baum-Welch training) may limit its ability to capture evolving patterns. Also, the use of fixed RSI thresholds and simple binning may lose finer-grained signals in the data. The model currently considers only SPY, which restricts its generalizability.

4.2. Potential Improvements

Future work could integrate semi-supervised learning to fine-tune emission probabilities based on recent data. A more adaptive transition model could be introduced using rolling windows or reinforcement learning techniques. Additionally, adding macroeconomic features such as interest rates or inflation could enhance regime accuracy. Evaluating performance across multiple indices like QQQ, IWM, or DIA would help assess how well the model gen- eralizes to other market instruments. Lastly, introducing regime duration modeling using Hidden semi-Markov Models could better reflect varying market phase lengths.